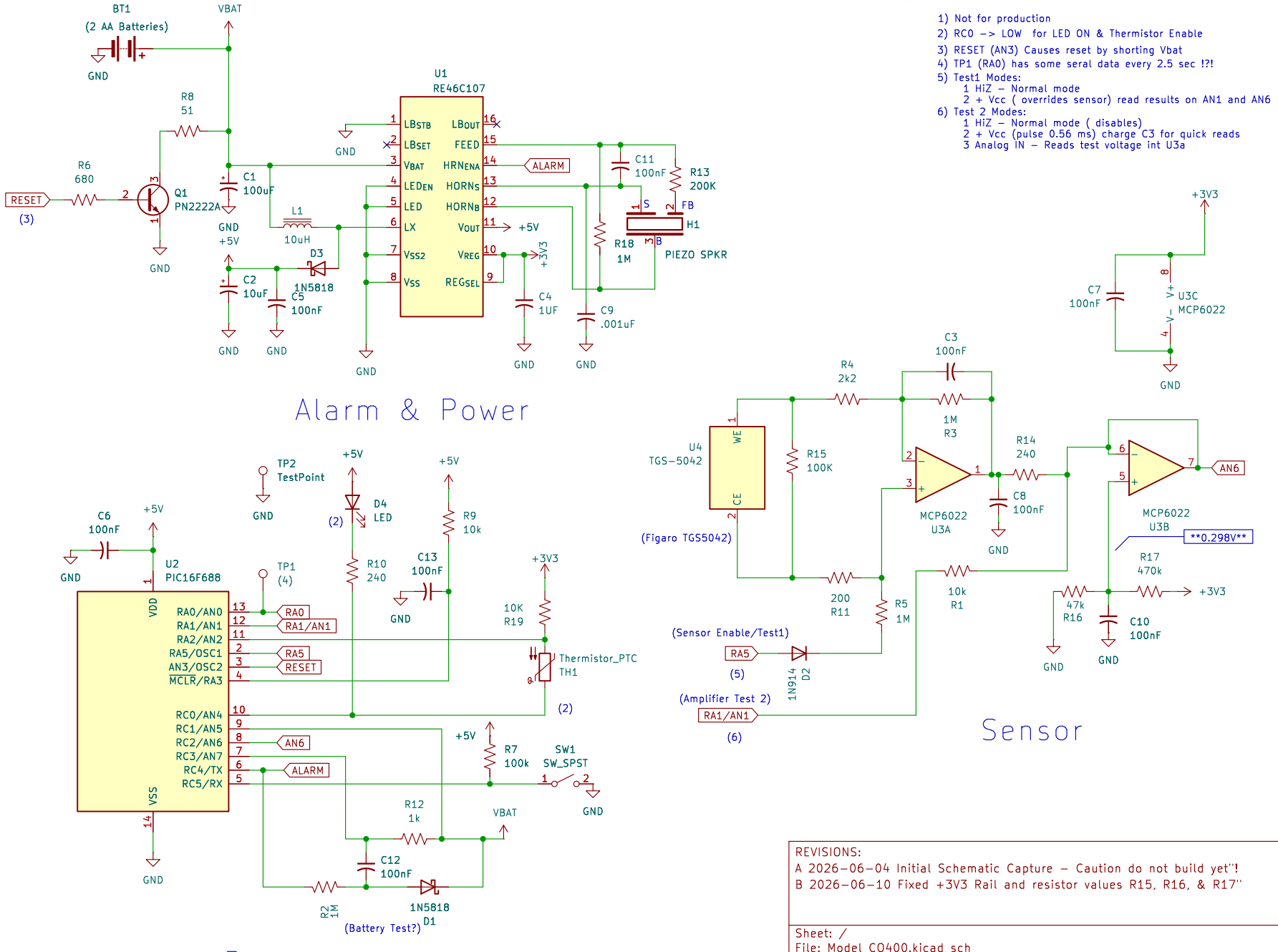
NOTES:
1) Not for production
2) RC0 –> LOW for LED ON & Thermistor Enable
3) RESET (AN3) Causes reset by shorting Vbat
4) TP1 (RA0) has some seral data every 2.5 sec !?!
5) Test1 Modes:
1 HiZ – Normal mode
2 + Vcc (overrides sensor) read results on AN1 and AN6
6) Test 2 Modes:
1 HiZ – Normal mode (disables)
2 + Vcc (pulse 0.56 ms) charge C3 for quick reads
3 Analog IN – Reads test voltage int U3a
BT1
(2 AA Batteries)
VBAT
GND
R8
51
R6
680
RESET
(3)
Q1
PN2222A
C1
100uF
L1
10uH
D3
1N5818
C2
10uF
C5
100nF
U1
RE46C107
LBSTB
LBSET
VBAT
LEDEN
LED
LX
VSS2
VSS
LBOUT
FEED
HRNENA
HORNS
HORNB
VOUT
VREG
REGSEL
ALARM
+5V
+3V3
C11
100nF
R13
200K
R18
1M
H1
PIEZO SPKR
C4
1UF
C9
.001uF
Alarm & Power
TP2
TestPoint
D4
LED
(2)
R9
10k
C13
100nF
C6
100nF
U2
PIC16F688
TP1
(4)
R10
240
10K
R19
Thermistor_PTC
TH1
RA0/AN0
RA1/AN1
RA2/AN2
RA5/OSC1
AN3/OSC2
MCLR/RA3
RC0/AN4
RC1/AN5
RC2/AN6
RC3/AN7
RC4/TX
RC5/RX
R7
100k
SW1
SW_SPST
R12
1k
C12
100nF
R2
1M
1N5818
D1
(Battery Test?)
Processor
U4
TGS-5042
WE
CE
(Figaro TGS5042)
R4
2k2
C3
100nF
1M
R3
R15
100K
MCP6022
U3A
R14
240
C8
100nF
200
R11
R5
1M
10k
R1
(Sensor Enable/Test1)
RA5
(5)
1N914
D2
(Amplifier Test 2)
RA1/AN1
(6)
C7
100nF
U3C
MCP6022
MCP6022
U3B
AN6
0.298V
R17
470k
47k
R16
C10
100nF
Sensor
REVISIONS:
A 2026-06-04 Initial Schematic Capture – Caution do not build yet"!
B 2026-06-10 Fixed +3V3 Rail and resistor values R15, R16, & R17"
Sheet: /
File: Model_CO400.kicad_sch
Title: First Alert Model CO400 CO Alarm (ca. 2009)
Size: A
Date: 2026-06-10
Rev: B
KiCad E.D.A. 9.0.1
Id: 1/1